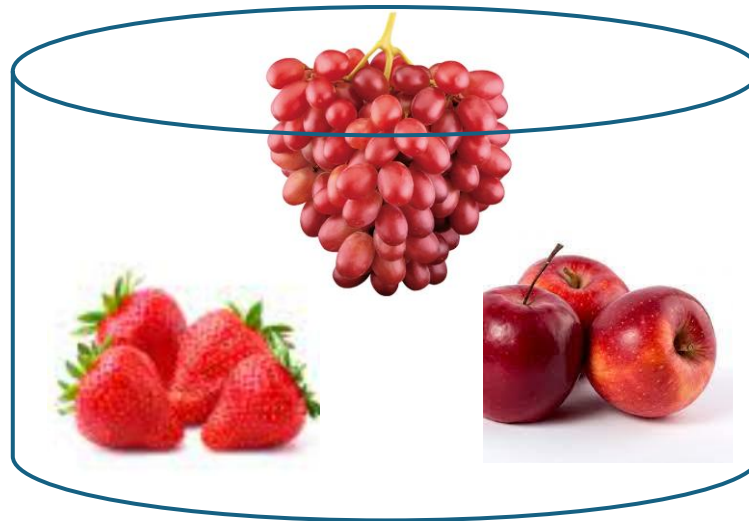
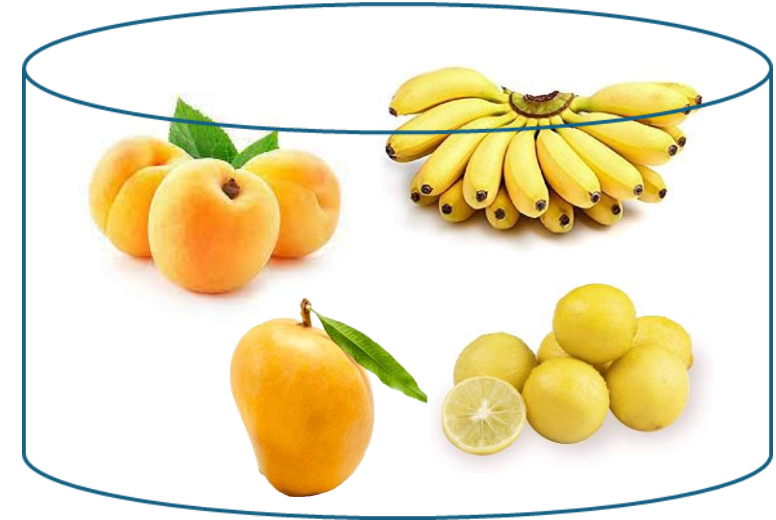
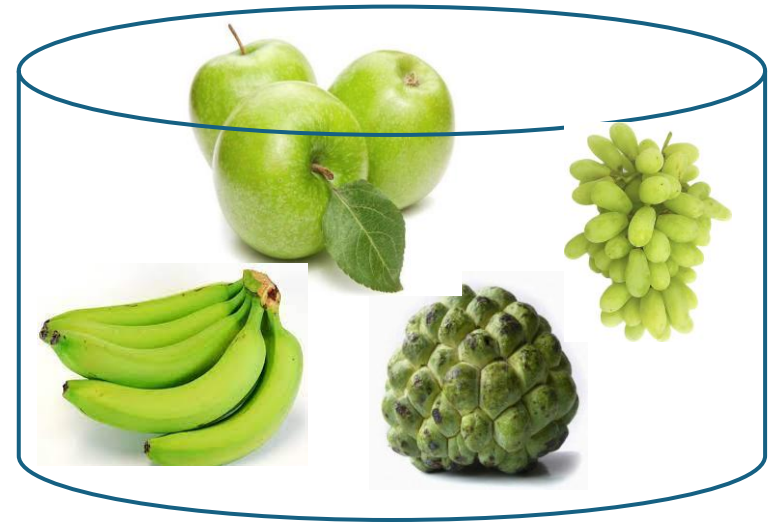


CLUSTERING ANALYSIS

- Importance of cluster analysis,
- K-means: The Basic K-means Algorithm,
- K-means: Additional Issues,
- K-means and Different Types of Clusters,
- Strengths and Weaknesses.

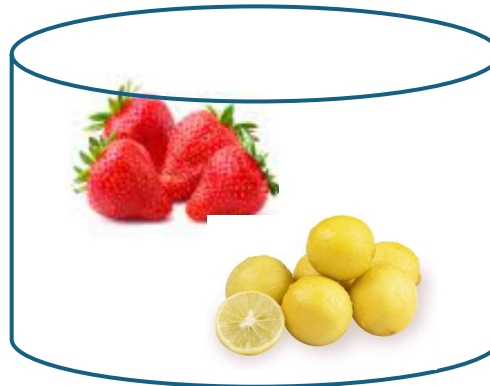
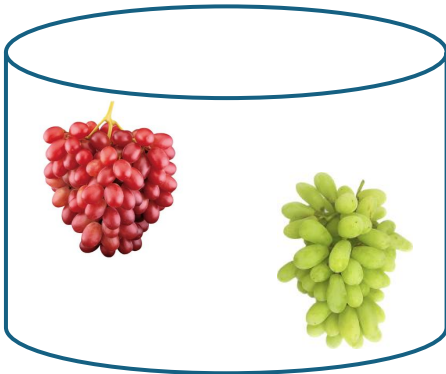
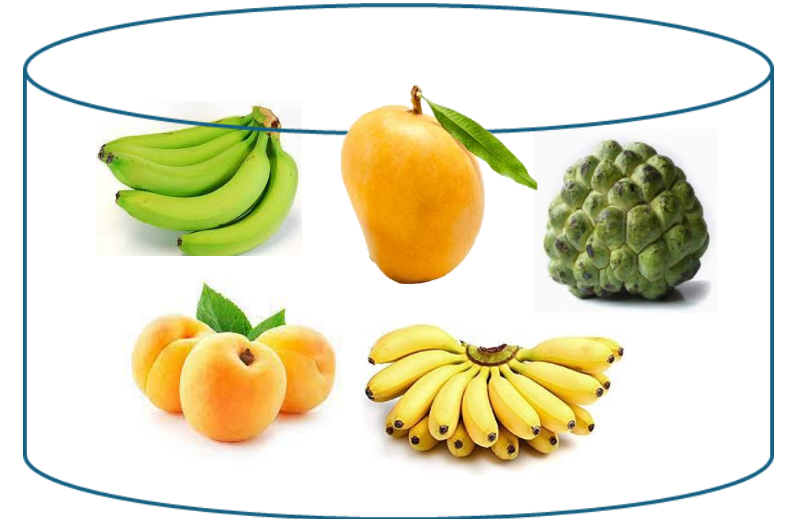
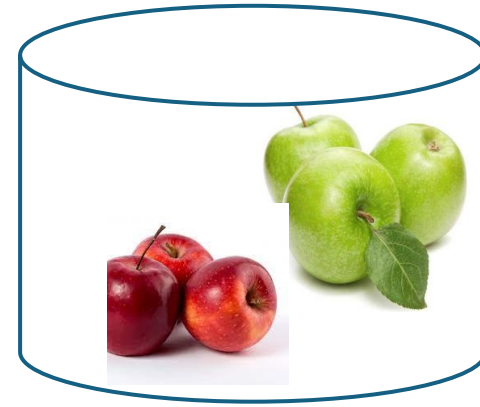


**CLUSTERING
ALGORITHM**





**CLUSTERING
ALGORITHM**



What is Clustering?

- Cluster analysis groups data objects based only on information found in the data that **describes the objects and their relationships**.
- The goal is that the **objects within a group be similar** (or related) to one another and **different from (or unrelated to) the objects in other groups**.
- The greater the similarity (or homogeneity) within a group and the greater the difference between groups, the better or more distinct the clustering.

- Because a cluster is a collection of data objects that are similar to one another within the cluster and dissimilar to objects in other clusters, a cluster of data objects can be treated as an implicit class.
 - In this sense, clustering is sometimes called **automatic classification**. This is a distinct advantage of cluster analysis.
- Clustering is also called **data segmentation** in some applications because clustering partitions large data sets into groups according to their similarity.
- Clustering can also be used for outlier detection, where outliers are values that are “far away” from any cluster.

Requirements of clustering in data mining

- **Scalability**: Many clustering algorithms *work well on small data* sets containing fewer than several hundred data objects.
- Clustering on only a **sample** of a given **large data** set may lead to **biased results**. Therefore, highly scalable clustering algorithms are needed.
- **Ability to deal with different types of attributes**: Many algorithms are designed to cluster **numeric** (interval-based) data. However, applications may require clustering other data types, such as binary, nominal (categorical), and ordinal data, or mixtures of these data types.

- **Discovery of clusters with arbitrary shape:** Many clustering algorithms determine clusters based on Euclidean or Manhattan distance measures.
 - Algorithms based on such distance measures tend to find spherical clusters with similar size and density.
- **However, a cluster could be of any shape.**
- Example: Consider **sensors**, deployed for **environment surveillance**.
 - Cluster analysis on sensor readings can detect interesting phenomena.
 - We may want to use clustering to find the frontier of a running forest fire, which is often not spherical.

- **Requirements for domain knowledge to determine input parameters:**
-
- Many clustering algorithms **require users to provide** domain knowledge in the form of **input parameters** such as the desired number of clusters.
- Consequently, the **clustering** results may be **sensitive** to such parameters.
- **Parameters are often hard to determine**, especially for **high-dimensionality data sets** and where users have yet to grasp a deep understanding of their data.

- **Ability to deal with noisy data:**
- Most **real-world data sets contain outliers** and/or missing, unknown, or erroneous data.
- **Sensor readings**, for example, are often noisy—some readings may be inaccurate due to the sensing mechanisms, and some readings may be **erroneous due to interferences from surrounding transient objects**.
- **Clustering algorithms can be sensitive** to such noise and may produce **poor-quality clusters**. Therefore, we need clustering methods that are robust to noise.

- **Incremental clustering and insensitivity to input order:**
- In many applications, incremental updates (representing newer data) may arrive at any time.
- Some clustering algorithms cannot incorporate incremental updates into existing clustering structures and, instead, have to recompute a new clustering from scratch.
- Clustering algorithms may also be sensitive to the input data order. That is, given a set of data objects, clustering algorithms may return dramatically different clustering depending on the order in which the objects are presented.

- **Capability of clustering high-dimensionality data:**

- A data set can contain numerous dimensions or attributes. When clustering documents, for example, each keyword can be regarded as a dimension, and there are often thousands of keywords.
- Most clustering algorithms are good at handling low-dimensional data such as data sets involving only two or three dimensions. Finding clusters of data objects in a high-dimensional space is challenging, especially considering that such data can be very sparse and highly skewed.

- **Constraint-based clustering:**

- Real-world applications may need to perform clustering under various kinds of constraints.
- Suppose that your job is to choose the locations for a given number of new automatic teller machines (ATMs) in a city.
- To decide upon this, you may cluster households while considering constraints such as the city's rivers and highway networks and the types and number of customers per cluster.
- A challenging task is to find data groups with good clustering behavior that satisfy specified constraints.

- **Interpretability and usability:**

- Users want clustering results to be interpretable, comprehensible, and usable.
- That is, clustering may need to be tied in with specific semantic interpretations and applications.
- It is important to study how an application goal may influence the selection of clustering features and clustering methods.

Comparing Aspects of Clustering Algorithm

- **The partitioning criteria:**
- In some methods, all the **objects are partitioned so that no hierarchy exists among the clusters**. That is, all the clusters are at the same level conceptually.
- Alternatively, other methods **partition data objects hierarchically**, where clusters can be formed at **different semantic levels**.
- For example, in text mining, we may want to organize a corpus of documents into multiple general topics, such as “**politics**” and “**sports**,” each of which may have subtopics, For instance, “football,” “basketball,” “baseball,” and “hockey” can exist as subtopics of “sports.”
 - The latter four subtopics are at a lower level in the hierarchy than “sports.”

- **Separation of clusters:**

- Some methods partition data objects into mutually exclusive clusters.
- When clustering customers into groups so that each group is taken care of by one manager, each customer may belong to only one group.
- In some other situations, the clusters may not be exclusive, that is, a data object may belong to more than one cluster.
- For example, when clustering documents into topics, a document may be related to multiple topics.

- **Similarity measure:**
- Some methods determine the **similarity between two objects by the distance between them**. Such a distance can be defined on Euclidean space, a road network, a vector space, or any other space.
- In other methods, the similarity may be defined by connectivity based on density or contiguity, and may not rely on the absolute distance between two objects.
- **Similarity measures play a fundamental role in the design of clustering methods.**
- While distance-based methods can often take advantage of optimization techniques, density- and continuity-based methods can often find clusters of arbitrary shape.

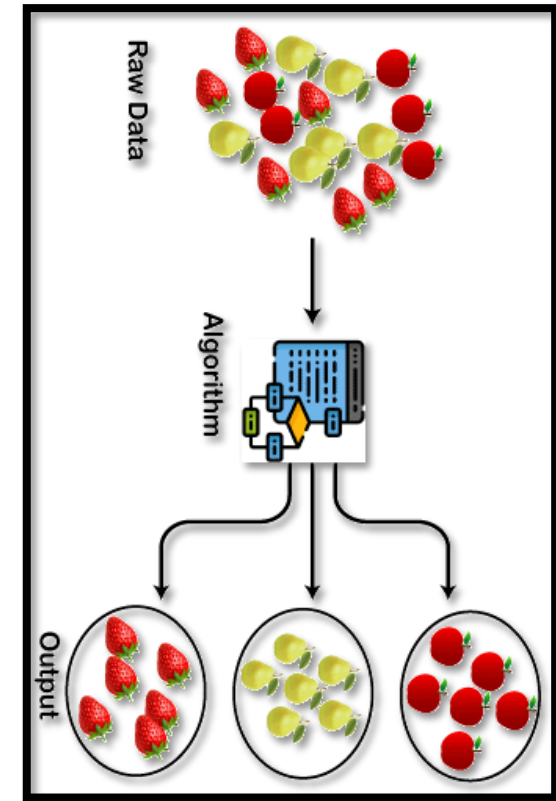
- **Clustering space:**

- Many clustering methods search for clusters within the entire given data space. These methods are **useful for low-dimensionality data sets**.
- With **high-dimensional data**, however, there can be many **irrelevant attributes**, which can make similarity measurements unreliable and clusters meaningless.

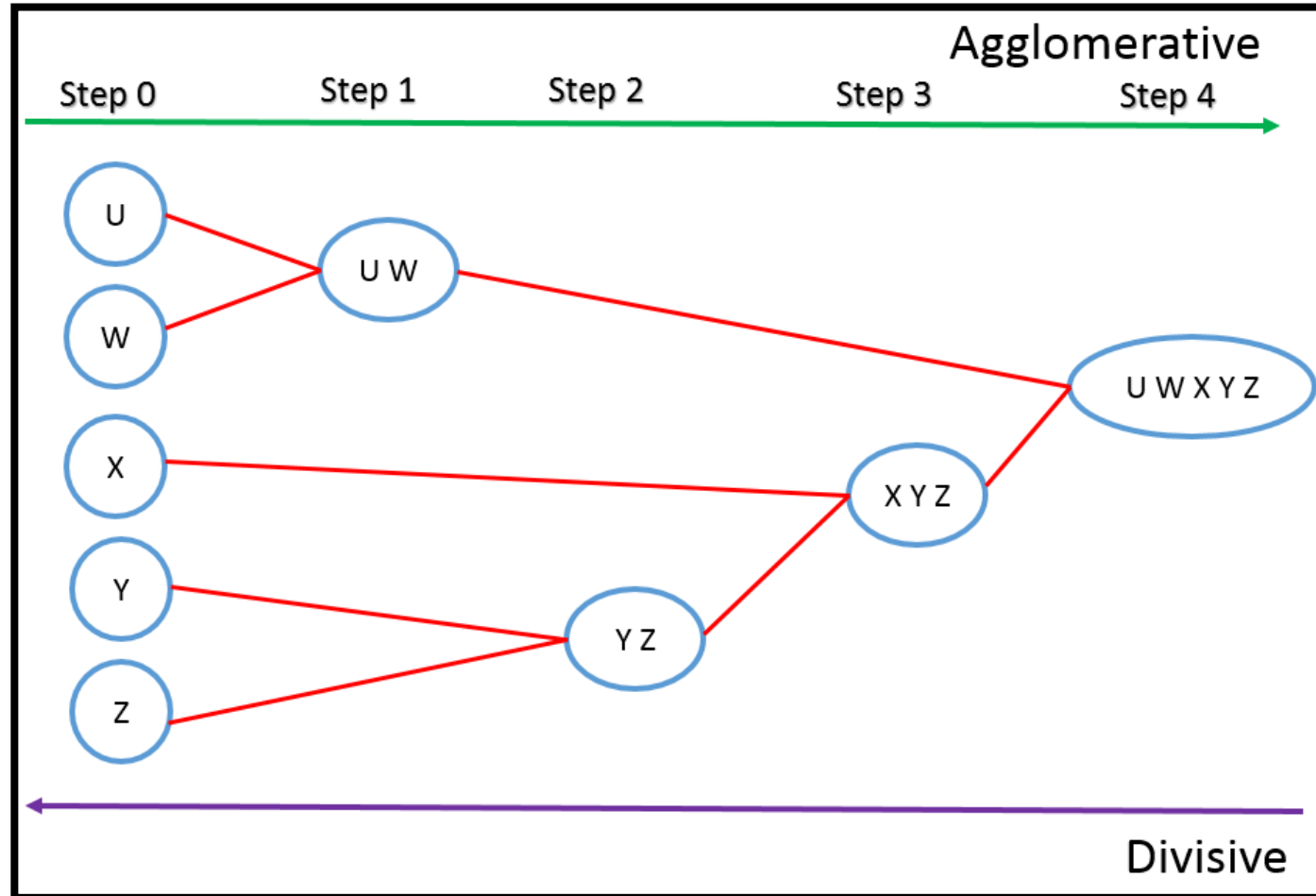
Types of Clustering Methods

- **Partitioning Methods**

- It is a type of clustering that **divides the data into non-hierarchical groups**. It is also known as the **centroid-based method**.
- Given a dataset of ‘n’ objects, a partitioning method assembles k partitions of the information, where each partition defines a cluster. ‘k’ < ‘n’.
 - Each group must include a minimum of one object.
 - Each object should apply to accurately one group.
- The general criterion of good partitioning is that objects in the same cluster are “close” or associated with each other, whereas objects of different clusters are “far apart” or very different.
- Example: *K-Means Clustering Algorithm*



Hierarchical Clustering



Hierarchical Clustering

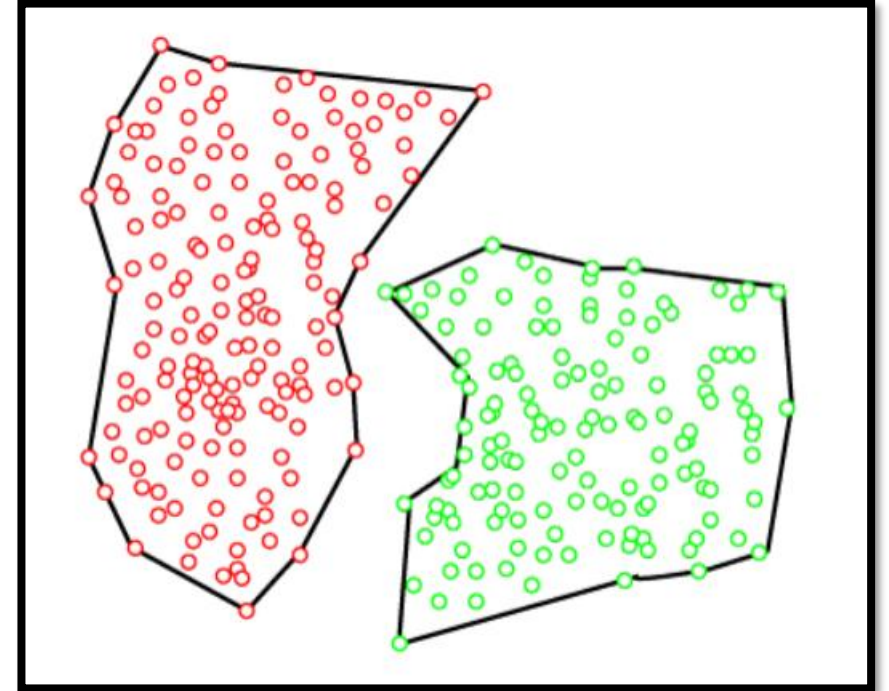
- Hierarchical clustering can be used as an **alternative for the partitioned** clustering as there is **no requirement of pre-specifying the number of clusters to be created**.
- In this technique, the dataset is divided into clusters to create a tree-like structure, which is also called a **dendrogram**.
- The observations or any number of clusters can be selected by cutting the tree at the correct level.
- A hierarchical method creates a hierarchical decomposition of the given set of data objects. A hierarchical method can be classified as being either **agglomerative or divisive**, based on how the hierarchical decomposition is formed.

Hierarchical Clustering

- The **agglomerative approach**, also called the **bottom-up approach**, starts with each object forming a separate group. It successively merges the objects or groups close to one another, until all the groups are merged into one (the topmost level of the hierarchy), or a termination condition holds.
- The **divisive approach**, also called the **top-down approach**, starts with all the objects in the same cluster. In each successive iteration, a cluster is split into smaller clusters, until eventually each object is in one cluster, or a termination condition holds.

Density-based methods

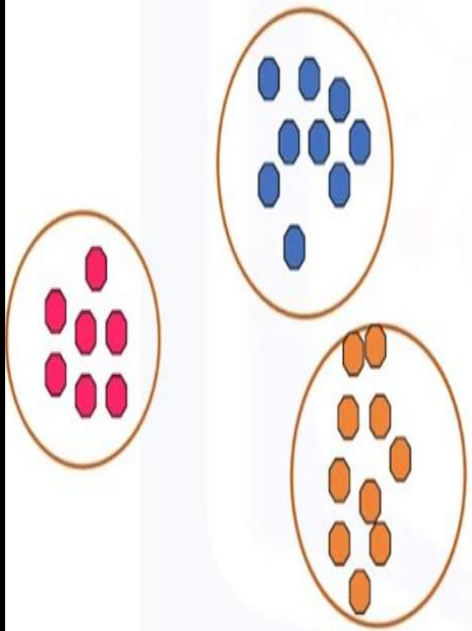
- Partitioning methods cluster objects based on the distance among objects. Such methods can discover *only spherical-shaped clusters* and encounter difficulty in finding clusters of arbitrary shapes.
- The density-based clustering method connects the highly-dense areas into clusters, and the arbitrarily shaped distributions are formed as long as the dense region can be connected.
- DBSCAN Algorithm Example



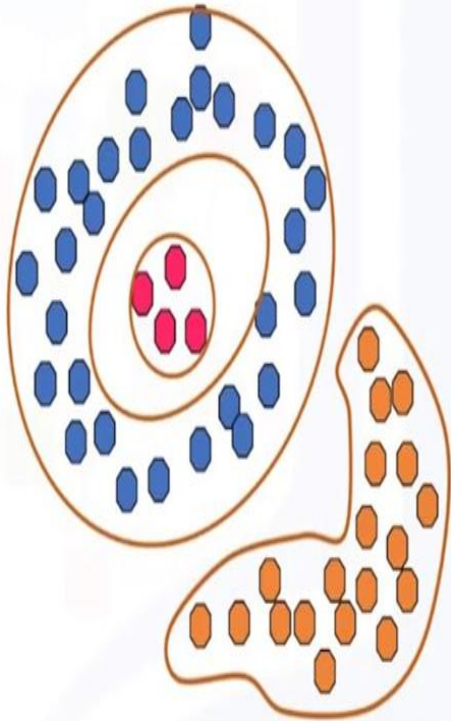
Density-based methods

- Their general idea is to **continue growing** a given cluster as long as the density (number of objects) in the “neighborhood” exceeds some threshold.
- For example, for each data point within a given cluster, the neighborhood of a given radius has to contain at least a minimum number of points.
- Such a method can be used to filter out noise or outliers and discover clusters of arbitrary shape.

• Spherical-shape clusters



• Arbitrary-shape clusters

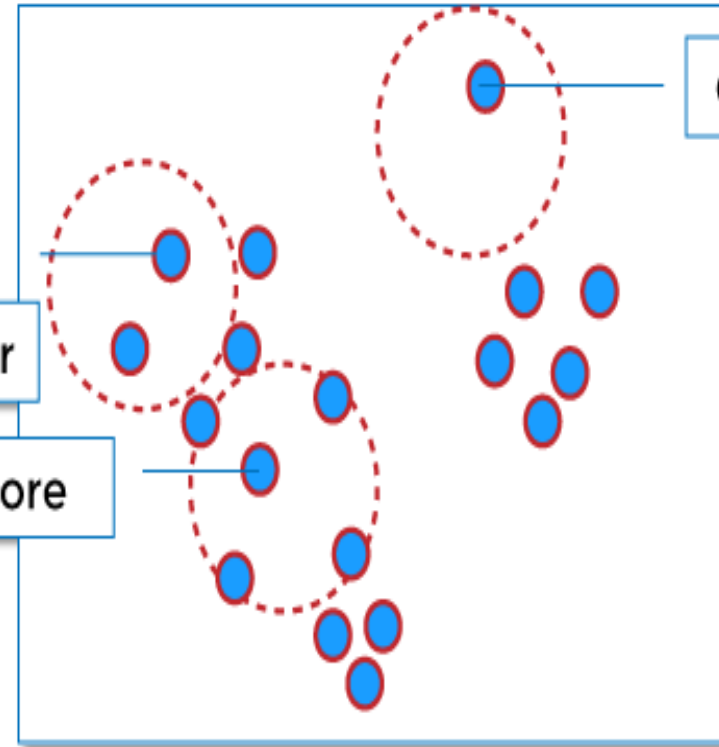


Border

Core

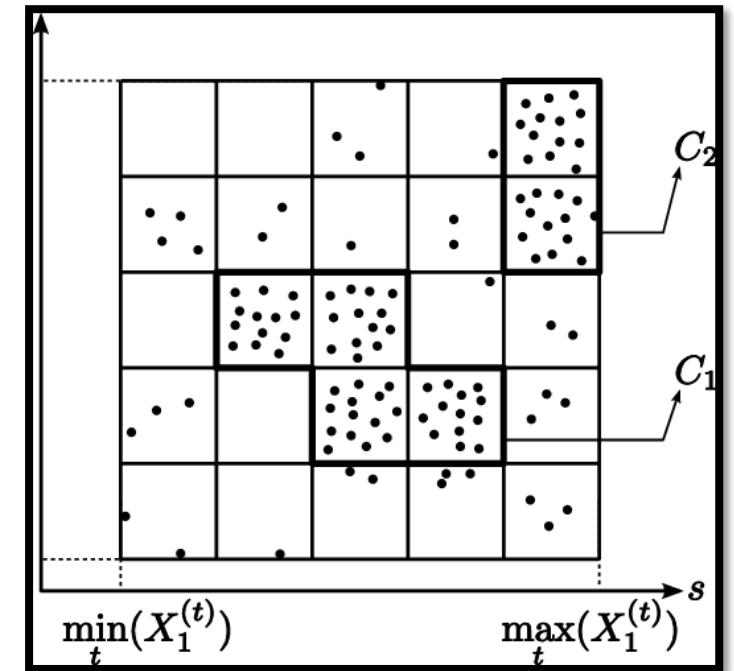
Outlier

Eps = 1cm
MinPts = 5



Grid Based Clustering Method

- Grid-based methods quantize the object space into a finite number of cells that form a grid structure.
- All the clustering operations are performed on the grid structure (i.e., on the quantized space).
- The main advantage of this approach is its **fast processing time**, which is typically independent of the number of data objects and dependent only on the number of cells in each dimension in the quantized space.



K-Means Clustering Algorithm

K-Means Clustering Algorithm

Algorithm 8.1 Basic K-means algorithm.

- 1: Select K points as initial centroids.
- 2: **repeat**
- 3: Form K clusters by assigning each point to its closest centroid.
- 4: Recompute the centroid of each cluster.
- 5: **until** Centroids do not change.

- Step 1 – First, we need to specify the number of clusters, K , need to be generated by this algorithm.
- Step 2 – Next, randomly select K data points and assign each data point to a cluster. In simple words, classify the data based on the number of data points.
- Step 3 – Now it will compute the cluster centroids.
- Step 4 – Next, keep iterating the following until we find optimal centroid which is the assignment of data points to the clusters that are not changing any more.
- K-means follows **Expectation-Maximization** approach to solve the problem. The Expectation-step is used for assigning the data points to the closest cluster and the Maximization-step is used for computing the centroid of each cluster.

a) Suppose that the data mining task is to cluster the following eight points with (x;y) representing location into three clusters. 09

A1(2;10); A2(2;5); A3(8;4); B1(5;8); B2(7;5); B3(6;4); C1(1;2); C2(4;9)

Suppose initially we assign A1, B1, and C1 as the center of each cluster, respectively. Use the k-means algorithm to show:

- i) The three cluster centers after two rounds of execution.
- ii) List the strengths and weakness of k-means
- iii) Write k-means algorithm

a) Write k – means Algorithms. Consider a two dimensional database D with the records: (9)

R1 (2,2) , R2 (2,4) R3(4,2), R4(4, 4), R5(3,6), R6(7,6), R7(9,6), R8(5, 10), R9(8, 10), R10(10, 10). Show the result of the k – means algorithm at each step, assuming that you start with two clusters (k=2) with centers C1= (6,6) and C2 = (9,7)

- b) Perform K-means clustering for the following data points into 3 clusters. The distance function is Euclidean distance. Initially assign P_2, P_4 and P_5 as center of each cluster respectively.

10

	x	y
P_1	2	7
P_2	2	5
P_3	8	5
P_4	5	6
P_5	7	10
P_6	6	3
P_7	1	2
P_8	4	9

- c) Discuss the strength and weakness of K-Means algorithm.

4

- c) Write the algorithm for basic K-means clustering. Also specify the advantages and disadvantages of K-means clustering.

08

- c) Group the given points into three clusters using K-means algorithm. Consider P_1, P_4 and P_7 as the initial cluster centroids and show the three clusters after two iterations.

Points	P_1	P_2	P_3	P_4	P_5	P_6	P_7	P_8
x	4	2	9	6	7	5	2	6
y	6	5	3	9	5	7	2	6

Euclidean Distance

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- Point 1 : (3,4,8,7)
- Point 2 : (7,6,2,9)
- $\sqrt{(3 - 7)^2 + (4 - 6)^2 + (8 - 2)^2 + (7 - 9)^2}$
- $\sqrt{(-4)^2 + (-2)^2 + (6)^2 + (-2)^2}$
- $\sqrt{16 + 4 + 36 + 4}$
- $\sqrt{60}$
- 7.74

Consider Dataset of 08 objects and perform cluster analysis using K-Means algorithm.

Point Name	Attribute 1	Attribute 2
A1	2	10
A2	2	5
A3	8	4
B1	5	8
B2	7	15
B3	6	4
C1	1	2
C2	4	9

Consider A1, B1 and C1 as the centre of the clusters respectively

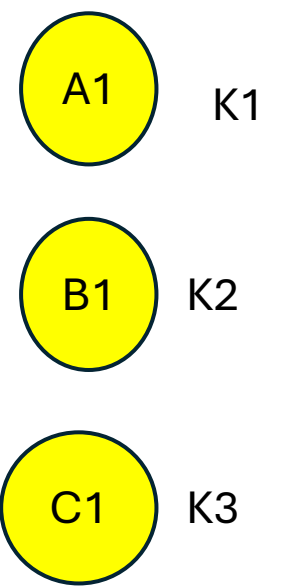
(x_1, y_1) is the coordinate of the first point

(x_2, y_2) is the coordinate of the second point.

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Point Name	Attribute 1	Attribute 2	Distance From		
			K1	K2	K3
A1	2	10	0		
A2	2	5			
A3	8	4			
B1	5	8		0	
B2	7	15			
B3	6	4			
C1	1	2			0
C2	4	9			

Iteration 1



Point Name	Attribute 1	Attribute 2	Distance From		
			K1	K2	K3
A1	2	10	0	3.60	8.06
A2	2	5			
A3	8	4			
B1	5	8			
B2	7	15			
B3	6	4			
C1	1	2			
C2	4	9			

Iteration 1

Point Name	Attribute 1	Attribute 2	Distance From		
			K1	K2	K3
A1	2	10	0	3.60	8.06
A2	2	5	5	4.24	3.16
A3	8	4	8.48	5	7.28
B1	5	8	3.60	0	7.21
B2	7	15	7.07	7.28	14.31
B3	6	4	7.21	4.12	5.38
C1	1	2	8.06	7.21	0
C2	4	9	2.23	1.41	7.61

Iteration 1

Point Name	Attribute 1	Attribute 2	Distance From			Allocated to
			K1	K2	K3	
A1	2	10	0	3.60	8.06	K1
A2	2	5	5	4.24	3.16	K3
A3	8	4	8.48	5	7.28	K2
B1	5	8	3.60	0	7.21	K2
B2	7	15	7.07	7.28	14.31	K1
B3	6	4	7.21	4.12	5.38	K2
C1	1	2	8.06	7.21	0	K3
C2	4	9	2.23	1.41	7.61	K2

Cluster Membership

K1	A1, B2
K2	B1, A3, B3, C2
K3	C1, A2

Iteration 1

Cluster Membership → New Mean

K1	A1, B2	$\{ (2 + 7) / 2 , (10 + 15) / 2 \}$
K2	B1, A3, B3, C2	$\{ (5 + 8 + 6 + 4) / 4 , (8 + 4 + 4 + 9) / 4 \}$
K3	C1, A2	$\{ (1 + 2) / 2 , (2 + 5) / 2 \}$

K1	A1, B2	(4.5 , 12.5)
K2	B1, A3, B3, C2	(5.75 , 6.25)
K3	C1, A2	(1.5 , 3.5)

Point Name	Attribute 1	Attribute 2	Distance From			Allocated to
			K1	K2	K3	
A1	2	10	3.53	5.30	6.51	K1
A2	2	5	7.90	3.95	1.58	K3
A3	8	4	9.19	3.18	6.51	K2
B1	5	8	4.52	1.90	5.70	K2
B2	7	15	3.53	8.83	12.74	K1
B3	6	4	8.63	2.26	4.52	K2
C1	1	2	11.06	6.37	1.58	K3
C2	4	9	3.53	3.25	6.04	K2

Iteration 2

Cluster Membership

K1	A1, B2
K2	B1, A3, B3, C2
K3	C1, A2

As the cluster membership does not change, K-Means iteration ends

Perform K-Means
Clustering on the
dataset. K=3

Point Name	Attribute 1	Attribute 2
A1	2	10
A2	2	6
A3	11	11
A4	6	9
A5	6	4
A6	1	2
A7	5	10
A8	4	9
A9	10	12
A10	7	5
A11	9	11
A12	4	6
A13	3	10
A14	3	8
A15	6	11

Iteration 1

K1	A2 , A5 , A6 , A10 , A12 , A14	(3.833, 5.167)
K2	A1 , A4 , A7, A8 , A13	(4, 9.6)
K3	A3 , A9, A11, A15	(9, 11.25)

Iteration 2

K1	A2 , A5 , A6 , A10 , A12	(4, 4.6)
K2	A1 , A4 , A7, A8 , A13 , A14, A15	(4.143, 9.571)
K3	A3 , A9, A11	(10, 11.333)

Iteration 3

K1	A2 , A5 , A6 , A10 , A12
K2	A1 , A4 , A7, A8 , A13 , A14, A15
K3	A3 , A9, A11

Advantages – K-Means

- 1. **Efficiency**
- K-Means clustering is known for its efficiency.
- It has a linear time complexity, which means that it can handle large datasets conveniently.
- With unlabeled big data, K-Means offers many insights and benefits as an unsupervised clustering algorithm.
- It can also be used with parallel computing to speed up the process, making it a faster and more efficient algorithm.

Advantages – K-Means

- 2. **Simplicity**
- One of the main advantages of K-Means clustering is its simplicity.
- It is relatively easy to implement and identify unknown groups of data from complex datasets.
- The results are presented in an easy and simple manner, which makes it easy to explain the results.
- With K-Means clustering, you can quickly get insights into your data.

Advantages – K-Means

- 3. Flexibility
- If there are any problems, adjusting the cluster segment will allow changes to easily occur on the algorithm.
- This flexibility makes K-Means clustering an ideal choice for data segmentation and pattern recognition.
- It can also be used with different distance metrics and initialization methods, which makes it a versatile algorithm that can be used for a wide range of applications.

Disadvantages of K-Means

- 1. **Sensitivity to Outliers**
- K-Means clustering is sensitive to outliers. Outliers are data points that are significantly different from other data points in the dataset.
- In K-Means clustering, outliers can distort the cluster centroids, leading to inaccurate clustering results. Therefore, it is important to preprocess the data and remove any outliers before applying K-Means clustering.

Disadvantages of K-Means

- 2. **Dependence on Initialization**
- K-Means clustering is dependent on the initialization of the cluster centroids. The initial positions of the centroids can significantly affect the final clustering results.
- If the initial positions of the centroids are not chosen carefully, K-Means clustering may converge to a local minimum instead of the global minimum, resulting in suboptimal clustering results. Therefore, it is important to choose the initial positions of the centroids carefully.

Disadvantages of K-Means

- 3. **Limitations with Cluster Shapes**
- K-Means clustering assumes that the clusters are spherical and have the same variance. However, in real-world datasets, the clusters may have different shapes and variances.

Agglomerative Hierarchical Clustering

- An **agglomerative hierarchical clustering** method uses a **bottom-up** strategy.
- It typically starts by letting each object form its own cluster and iteratively merges clusters into larger and larger clusters, until all the objects are in a single cluster or certain termination conditions are satisfied.
- The single cluster becomes the hierarchy's root.

Agglomerative Hierarchical Clustering

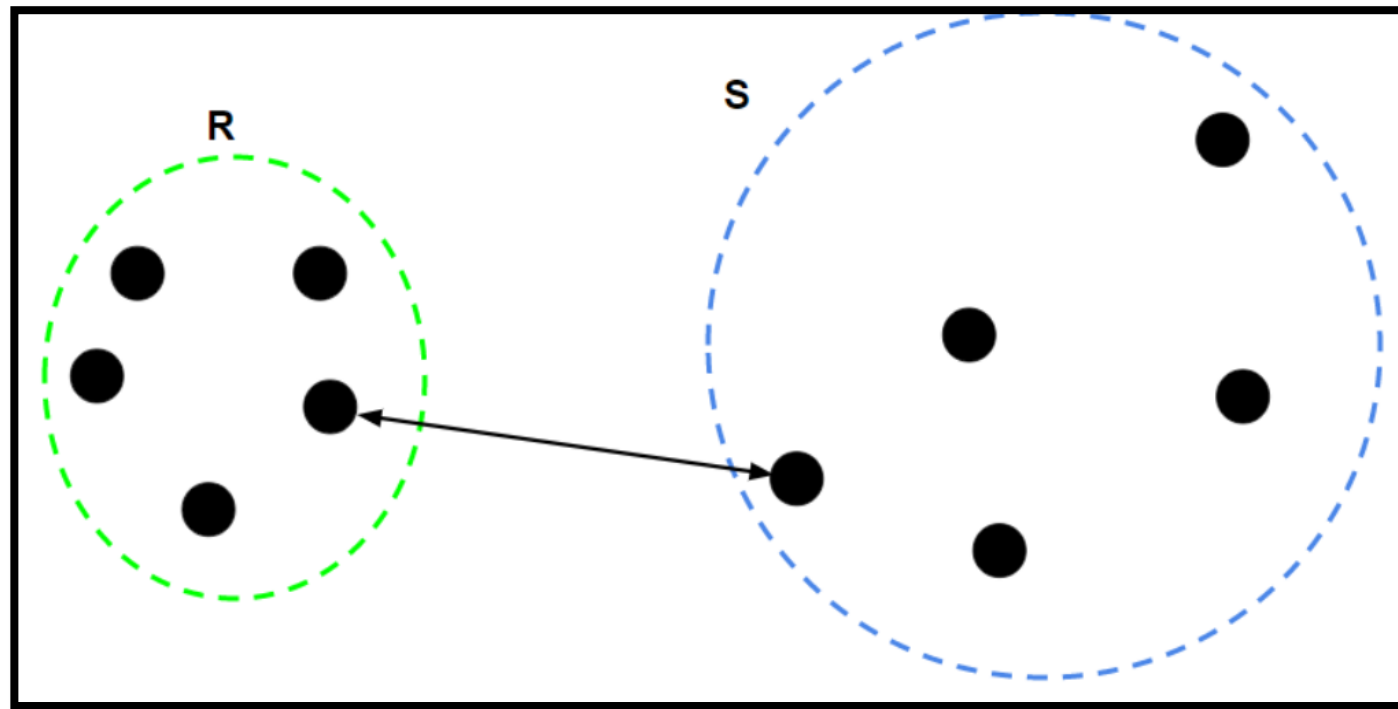
- For the merging step, it finds the two clusters that are closest to each other, according to similarity measure and combines the two to form one cluster.
- Because two clusters are merged per iteration, where each cluster contains at least one object, an **agglomerative method requires at most n iterations**.

Divisive Hierarchical Clustering

- A divisive hierarchical clustering method employs a top-down strategy.
- It starts by placing all objects in one cluster, which is the hierarchy's root.
- It then divides the root cluster into several smaller subclusters, and recursively partitions those clusters into smaller ones.
- The partitioning process continues until each cluster at the lowest level is coherent enough—either containing only one object, or the objects within a cluster are sufficiently similar to each other.

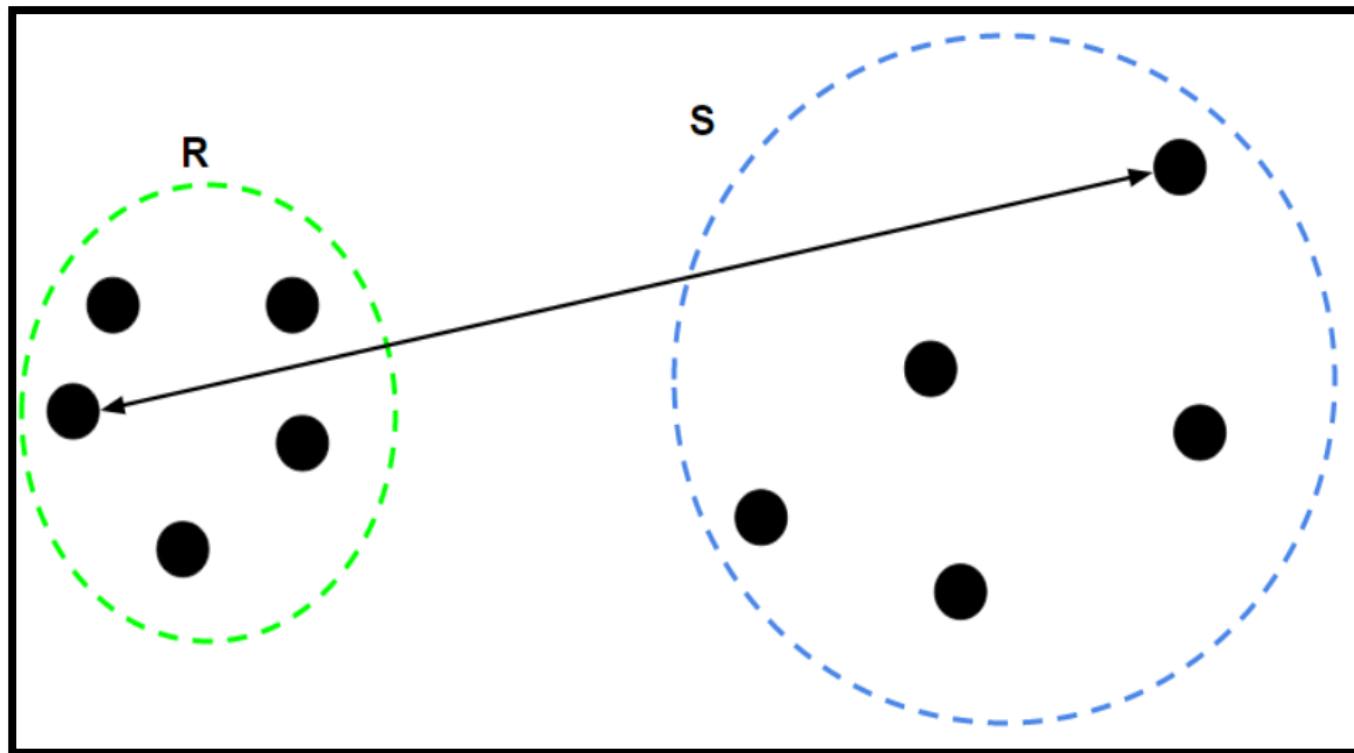
Single Linkage

- For two clusters R and S, the single linkage returns the **minimum distance between two points i and j** such that i belongs to R and j belongs to S.



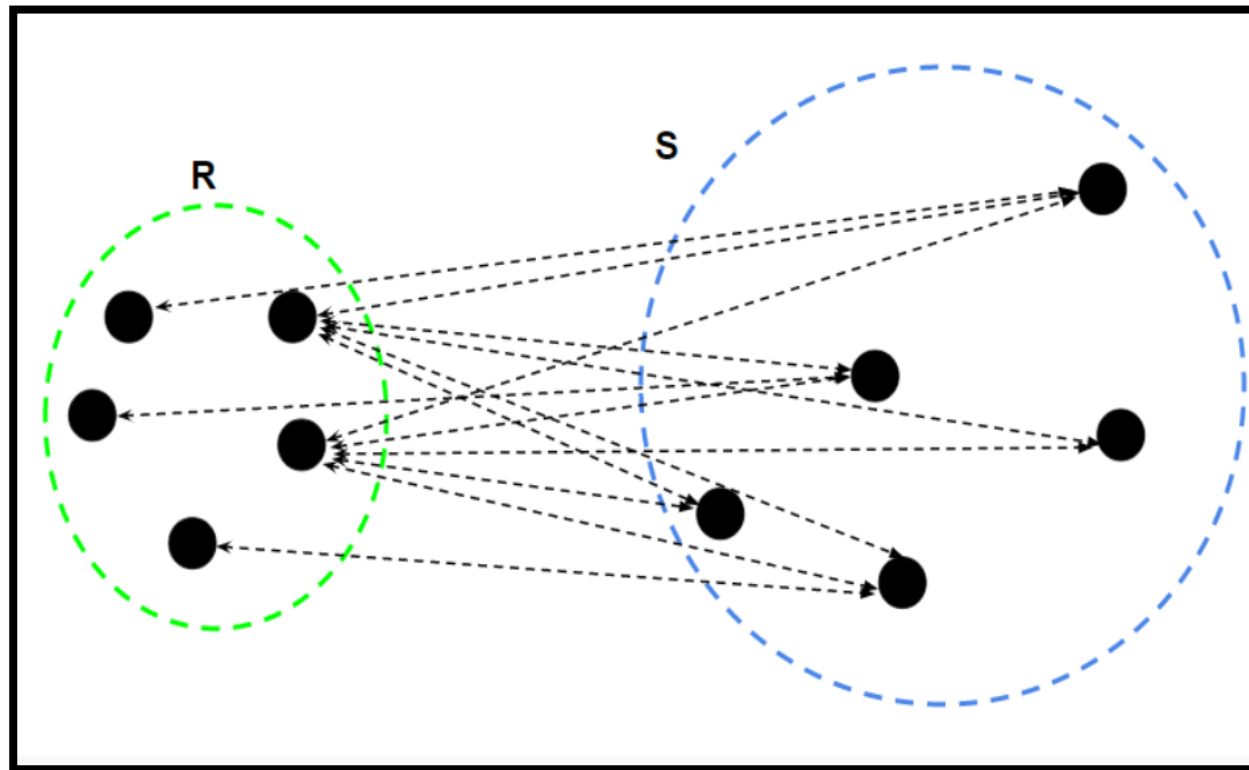
Complete Linkage

- For two clusters R and S, the complete linkage returns the maximum distance between two points i and j such that i belongs to R and j belongs to S.

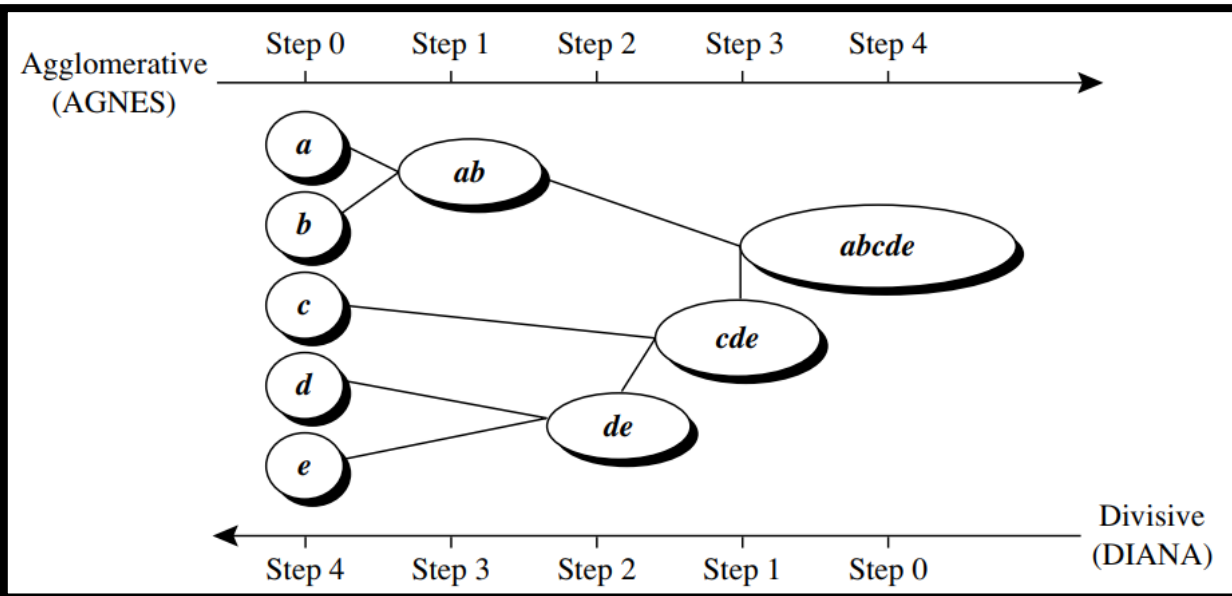


Average Linkage

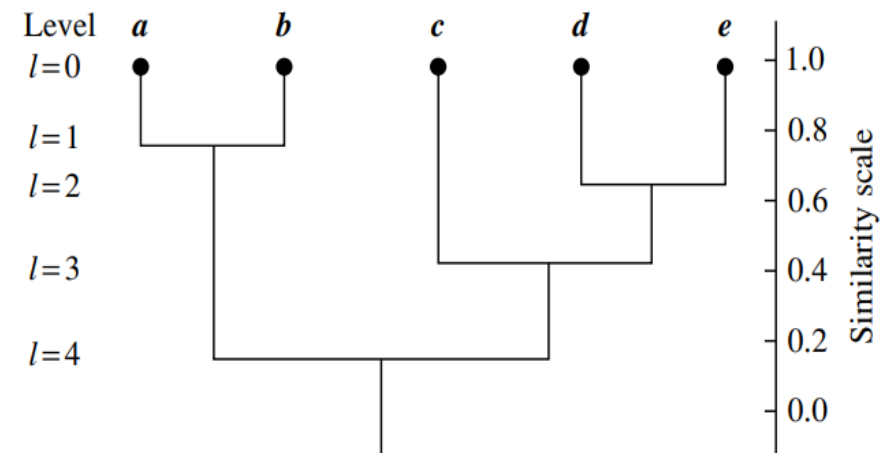
- For two clusters R and S, first for the distance between any data-point i in R and any data-point j in S and then the arithmetic mean of these distances are calculated. Average Linkage returns this value of the arithmetic mean.



AGNES (AGglomerative NESting)



Agglomerative and divisive hierarchical clustering on data objects $\{a, b, c, d, e\}$.



Step 1: Compute single linkages for every data point

	A1	A2	A3	A4	A5	A6
A1	0					
A2	0.23	0				
A3	0.22	0.14	0			
A4	0.37	0.19	0.13	0		
A5	0.34	0.14	0.28	0.23	0	
A6	0.24	0.24	0.10	0.22	0.39	0

Step 2: Find minimum distance between any data point.

	A1	A2	A3	A4	A5	A6
A1	0					
A2	0.23	0				
A3	0.22	0.14	0			
A4	0.37	0.19	0.13	0		
A5	0.34	0.14	0.28	0.23	0	
A6	0.24	0.24	0.10	0.22	0.39	0

Step 3: Form a cluster

	A1	A2	A3,A6	A4	A5
A1	0				
A2	0.23	0			
A3,A6	0.22	0.14	0		
A4	0.37	0.19	0.13	0	
A5	0.34	0.14	0.28	0.23	0

Competition between:
A1,A3 → 0.22 & A1,A6 → 0.24

Competition between:
A2,A3 → 0.14 & A2,A6 → 0.24

Competition between:
A4,A3 → 0.13 & A4,A6 → 0.22

Competition between:
A5,A3 → 0.28 & A5,A6 → 0.39

{ {A3,A6}, }

Repeat Step 1 to 3 till no more clusters possible

	A1	A2	A3,A6	A4	A5
A1	0				
A2	0.23	0			
A3,A6	0.22	0.14	0		
A4	0.37	0.19	0.13	0	
A5	0.34	0.14	0.28	0.28	0

{ {A3,A6}, }

Repeat Step 1 to 3 till no more clusters possible

	A1	A2	A3,A6,A4	A5
A1	0			
A2	0.23	0		
A3,A6,A4	0.22	0.14	0	
A5	0.34	0.14	0.28	0

{ { {A3,A6}, A4 } , }

Repeat Step 1 to 3 till no more clusters possible

	A1	A2	A3,A6,A4	A5
A1	0			
A2	0.23	0		
A3,A6,A4	0.22	0.14	0	
A5	0.34	0.14	0.28	0

{ { {A3,A6}, A4 } , }

Repeat Step 1 to 3 till no more clusters possible

	A1	A2,A5	A3,A6,A4
A1	0		
A2,A5	0.23	0	
A3,A6,A4	0.22	0.14	0

{ { {A3,A6}, A4 } , {A2,A5} , }

Repeat Step 1 to 3 till no more clusters possible

	A1	A2,A5	A3,A6,A4
A1	0		
A2,A5	0.23	0	
A3,A6,A4	0.22	0.14	0

{ { {A3,A6}, A4 } , {A2,A5} , }

Repeat Step 1 to 3 till no more clusters possible

	A1	A2,A5, A3,A6,A4
A1	0	
A2,A5, A3,A6,A4	0.23	0

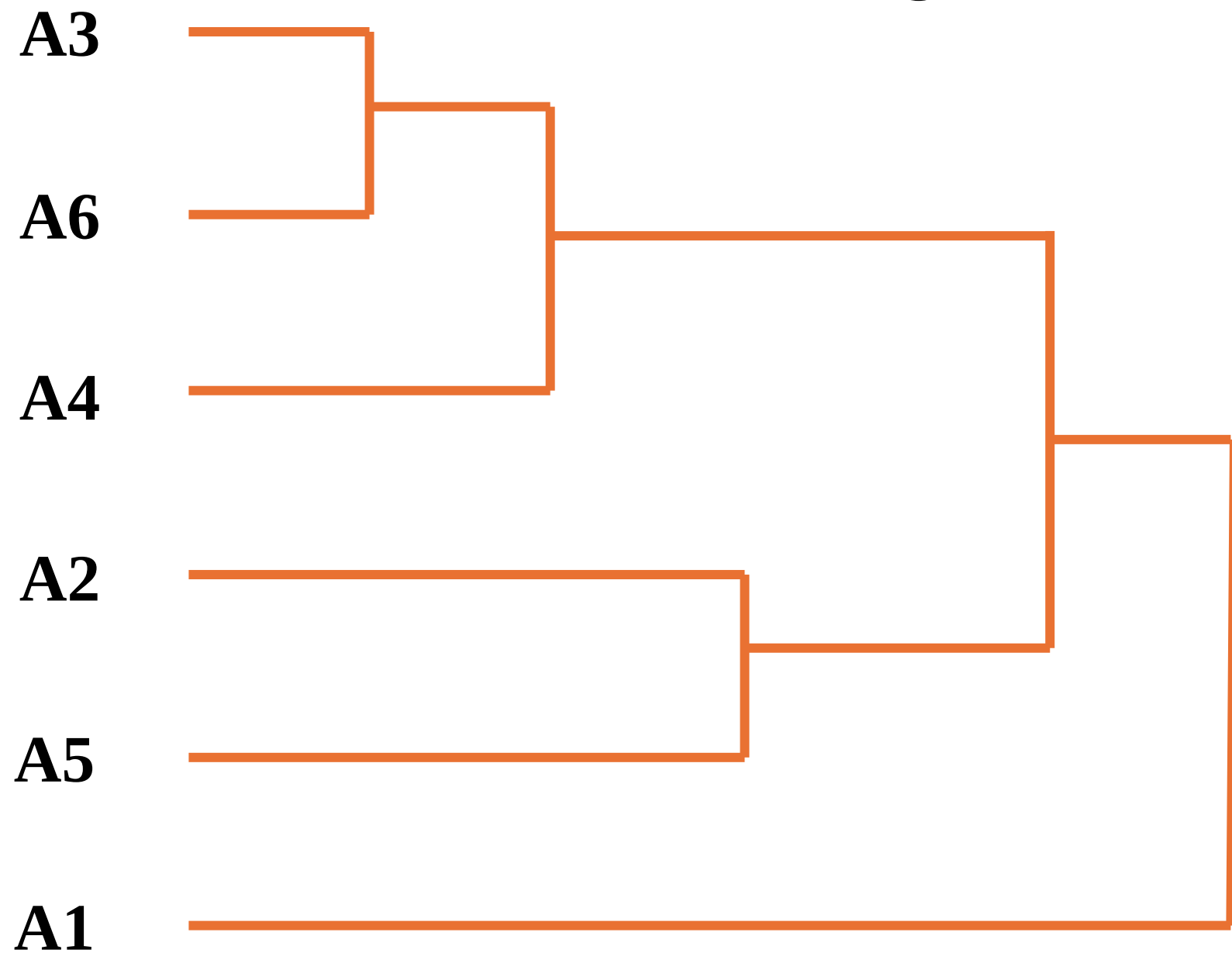
$\{ \{ \{A3,A6\}, A4 \}, \{A2,A5\} \}$

Only 01 minimum distance available –
directly merge to form cluster

	A1	A2,A5, A3,A6,A4
A1	0	
A2,A5, A3,A6,A4	0.23	0

$\{ \{ \{ \{A3,A6\}, A4 \}, \{A2,A5\} \}, A1 \}$

Dendrogram



Q2. Perform Agglomerative Hierarchical Clustering and illustrate results with a dendrogram using single linkage.

Point Name	x	y
A1	0.40	0.53
A2	0.22	0.38
A3	0.35	0.32
A4	0.26	0.19
A5	0.08	0.41
A6	0.45	0.30

Q3. Perform Agglomerative Hierarchical Clustering and illustrate results with a dendrogram using all types of linkage.

Point Name	Attribute 1
A1	1
A2	5
A3	8
A4	10
A5	2

Complete Linkage

	A1	A2	A3	A4	A5
A1	0				
A2		0			
A3			0		
A4				0	
A5					0

	A1	A2	A3	A4	A5
A1	0				
A2	4	0			
A3	7	3	0		
A4	9	5	2	0	
A5	1	3	6	8	0

Find the minimum distance between 2 points and form the cluster

	A1,A5	A2	A3	A4
A1,A5	0			
A2	4	0		
A3	7	3	0	
A4	9	5	2	0

Find the distance from newly formed cluster and remaining elements using COMPLETE LINKAGE FORMULA

	A1	A2	A3	A4	A5
A1	0				
A2	4	0			
A3	7	3	0		
A4	9	5	2	0	
A5	1	3	6	8	0

	A1,A5	A2	A3	A4
A1,A5	0			
A2	4	0		
A3	7	3	0	
A4	9	5	2	0

$$\text{distance}\{A2, \{A1, A5\}\} = \text{Max} (\text{dist}(A2, A1) , \text{dist}(A2, A5)) = \text{Max} (4, 3) = 4$$

$$\text{distance}\{A3, \{A1, A5\}\} = \text{Max} (\text{dist}(A3, A1) , \text{dist}(A3, A5)) = \text{Max} (7, 6) = 7$$

$$\text{distance}\{A4, \{A1, A5\}\} = \text{Max} (\text{dist}(A4, A1) , \text{dist}(A4, A5)) = \text{Max} (9, 8) = 9$$

Next Iteration Starts

	A1,A5	A2	A3	A4
A1,A5	0			
A2	4	0		
A3	7	3	0	
A4	9	5	2	0

Next Iteration Starts

	A1,A5	A2	A3,A4
A1,A5	0		
A2	4	0	
A3,A4	7	3	0

	A1,A5	A2	A3	A4
A1,A5	0			
A2	4	0		
A3	7	3	0	
A4	9	5	2	0

	A1,A5	A2	A3,A4
A1,A5	0		
A2	4	0	
A3,A4	9	5	0

$$\text{distance}\{\{ A1,A5 \} , \{ A3,A4 \} \} = \text{Max} (\text{dist}((A1,A5) , A3) , \text{dist}((A1,A5), A4)) \\ = \text{Max} (7, 9) = 9$$

$$\text{distance}\{ A2, \{ A3,A4 \} \} = \text{Max} (\text{dist}(A2,A3) , \text{dist}(A2,A4)) = \text{Max} (3, 5) = 5$$

Next Iteration starts

	A1,A5	A2	A3,A4
A1,A5	0		
A2	4	0	
A3,A4	9	5	0

Final Clusters

	(A1,A5), A2	A3,A4
(A1,A5), A2	0	
A3,A4	9	0

Advantages of Hierarchical clustering

- It is simple to implement and gives the best output in some cases.
- It is easy and results in a hierarchy, a structure that contains more information.
- It does not need us to pre-specify the number of clusters.

Disadvantages of Hierarchical clustering

- It breaks the large clusters.
- It is Difficult to handle different sized clusters and convex shapes.
- It is sensitive to noise and outliers.
- The algorithm can never be changed or deleted once it was done previously.

Key Issues in Hierarchical Clustering

- **Lack of a Global Objective Function:** agglomerative hierarchical clustering techniques perform clustering on a local level and as such there is no global objective function like in the K-Means algorithm. This is actually an advantage of this technique because the time and space complexity of global functions tends to be very expensive.
- **Ability to Handle Different cluster Sizes:** we have to decide how to treat clusters of various sizes that are merged together.
- **Merging Decisions Are Final:** one downside of this technique is that once two clusters have been merged they cannot be split up at a later time for a more favorable union.

Q4. Solve using Single, Complete and Average Linkage and comment on the formation of the clusters

	A1	A2	A3	A4	A5	A6
A1	0					
A2	5	0				
A3	14	9	0			
A4	11	2	13	0		
A5	18	15	6	3	0	
A6	10	16	8	10	11	0

Outlier Analysis

- Outlier detection (also known as anomaly detection) is the process of finding data objects with behaviors that are very different from expectation. Such objects are called outliers or anomalies.
- Outlier detection and clustering analysis are two highly related tasks. Clustering finds the majority patterns in a data set and organizes the data accordingly, whereas outlier detection tries to capture those exceptional cases that deviate substantially from the majority patterns.

What Are Outliers?

- An outlier is a data object that deviates significantly from the rest of the objects, as if it were generated by a different mechanism.
- Outliers are different from noisy data.
- Noise is a random error or variance in a measured variable.

TYPES OF OUTLIERS

Global Outlier

- **Definition:** Global outliers are data points that deviate significantly from the overall distribution of a dataset.
- **Causes:** Errors in data collection, measurement errors, or truly unusual events can result in global outliers.
- **Impact:** Global outliers can distort data analysis results and affect machine learning model performance.
- **Detection:** Techniques include statistical methods, machine learning algorithms and data visualization techniques.
- **Handling:** Options may include removing or correcting outliers, transforming data, or using robust methods.

Contextual / Conditional Outliers

- **Definition:** Contextual outliers are data points that deviate significantly from the expected behavior within a specific context or subgroup.
- **Characteristics:** Contextual outliers may not be outliers when considered in the entire dataset, but they exhibit unusual behavior within a specific context or subgroup.
- **Detection:** Techniques for detecting contextual outliers include contextual anomaly detection, and context-aware machine learning approaches.
- **Contextual Information:** such as time, location, or other relevant factors are crucial in identifying contextual outliers.
- **Handling:** Handling contextual outliers may involve considering the contextual information, contextual normalization or transformation of data, or using context-specific models or algorithms.

- **Contextual attributes:** The contextual attributes of a data object define the object's context. In the temperature example, the contextual attributes may be date and location.
- **Behavioral attributes:** These define the object's characteristics, and are used to evaluate whether the object is an outlier in the context to which it belongs. In the temperature example, the behavioral attributes may be the temperature, humidity, and pressure.

Collective Outliers

- **Definition:** Collective outliers are groups of data points that collectively deviate significantly from the overall distribution of a dataset.
- **Characteristics:** Collective outliers may not be outliers when considered individually, but as a group, they exhibit unusual behavior.
- **Detection:** Techniques for detecting collective outliers include clustering algorithms- density-based methods.
- **Impact:** Collective outliers can represent interesting patterns or anomalies in data that may require special attention or further investigation.
- **Handling:** Handling collective outliers depends on the specific use case and may involve further analysis of the group behavior, identification of contributing factors, or considering contextual information.
- Example - supply-chain delay

Challenges of Outlier Detection

- Modeling normal objects and outliers effectively.
- Application-specific outlier detection.
- Handling noise in outlier detection.
- Understandability.

Outlier Detection Methods

Supervised Methods

- Supervised methods model data normality and abnormality.
- Domain experts examine and label a sample of the underlying data.
- Outlier detection can then be modeled as a classification problem.
 - The two classes (i.e., normal objects versus outliers) are imbalanced.
 - That is, the population of outliers is typically much smaller than that of normal objects (**imbalanced classes**).
 - The lack of outlier samples can limit the capability of classifiers.
 - Find more number of outlier point and avoid mislabeling.

Unsupervised Methods

- Unsupervised outlier detection methods make an implicit assumption: **The normal objects are somewhat “clustered”**.
- In other words, an unsupervised outlier detection method expects that normal objects follow a pattern far more frequently than outliers.
- Unsupervised methods cannot detect outliers effectively. In some applications, normal objects are diversely distributed, and many such objects do not follow strong patterns.
- Ex: intrusion detection and computer virus detection problems, normal activities are very diverse.

Semi-Supervised Methods

- In several applications, although obtaining some labeled instance is possible, the number of such labeled instances is small.
- It can encounter cases where only a small group of the normal and outlier objects are labeled, but some data are unlabeled.
- Semi-supervised outlier detection methods were produced to tackle such methods.

- C) Define Outlier. Explain the issues faced by statistical approach for outlier detection. (5)
- c) What are the issues that need to be considered during anomaly detection? (6)
- c) Compare and contrast the different techniques for anomaly detection. (6)
- C) Write short note on: (7)
- a. Statistical Based Outlier Detection
 - b. Distance-Based Outlier Detection
- b) List and explain any two methods of outlier detection. 6
- C) What is an outlier? Explain in brief data Mining Applications (6)
- A) Write short note on: (12)
- a. Statistical Based Outlier Detection
 - b. Distance-Based Outlier Detection and
 - c. Deviation-Based Outlier Detection

Statistical Based Outlier Detection

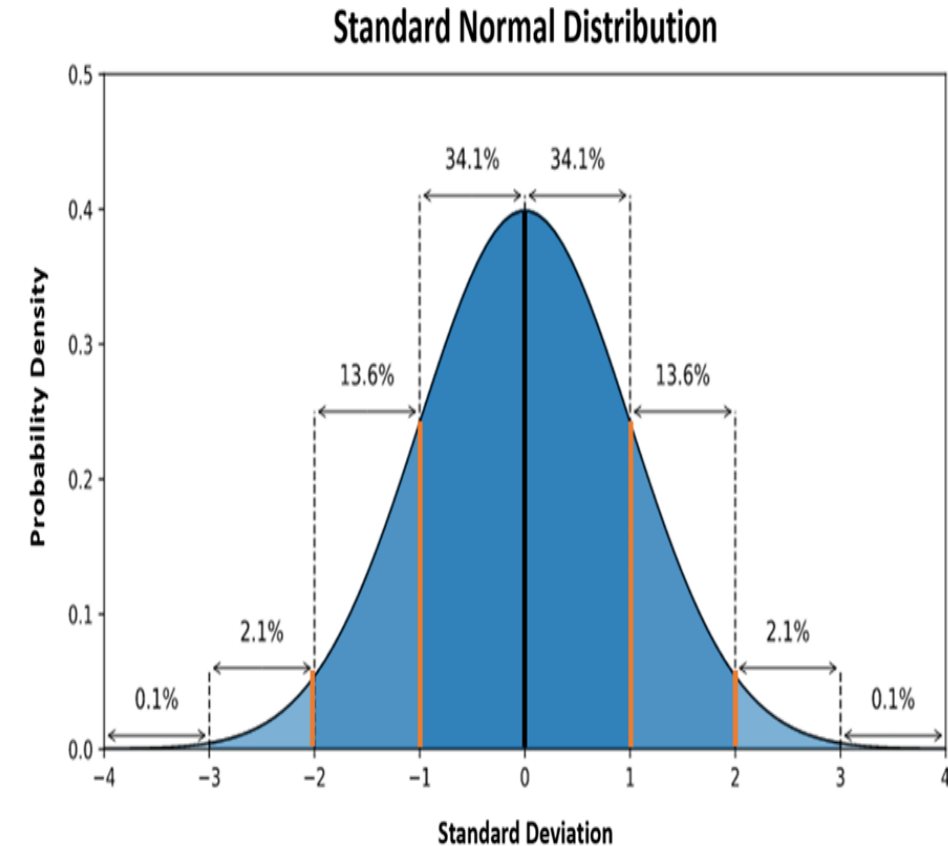
- Statistical methods for outlier detection rely on the assumption that **normal data objects** follow a specific **stochastic process**.
- We assume that normal data points occur in regions of **high probability**, while outliers appear in regions of **low probability**.
- **Normal data objects** refer to data points that conform to the expected pattern or distribution in a dataset. They follow the general trend and lie within the typical range of values, making them part of the majority group.
- For example, in a dataset of student test scores (out of 100), if most students score between 50 and 80, those scores are considered normal data objects. A score of 5 or 99 might be considered an outlier because it is unusually low or high.

Statistical Based Outlier Detection

- A **stochastic process** is a collection of random variables that evolve over time or space, following probabilistic rules. It describes a system where outcomes are uncertain but follow an underlying statistical pattern.
- **Examples of Stochastic Processes:**
 - **Stock Prices:** The price of a stock fluctuates randomly but follows market trends over time.
 - **Weather Patterns:** Temperature variations follow a probabilistic pattern but are unpredictable for specific days.

Statistical Based Outlier Detection

- **High-Probability Regions (Normal Data)**
- These are areas where most data points are concentrated, meaning the probability of a random data point falling in this region is **high**.
- Example: In a **height dataset** for adult males, most heights range between **5'5" and 6'2"**. This range is a **high-probability region**.
- **Low-Probability Regions (Outliers)**
- These are areas where data points are rarely found, meaning the probability of a random data point appearing here is **low**. Data points in this region are considered **outliers**.
- Example: In the same height dataset, someone who is **7 feet tall or 4 feet tall** would be in a **low-probability region**, making them an outlier.



Statistical Based Outlier Detection

- Statistical Approaches
 - Parametric Methods for Outlier Detection
 - Parametric methods assume that the data follows a known **probability distribution** (e.g., Gaussian, Poisson, or Exponential). These methods estimate the **parameters** of the distribution (such as mean and variance) and use statistical tests to detect outliers.
 - Non-Parametric Methods for Outlier Detection
 - Non-parametric methods **do not assume** an underlying probability distribution. Instead, they use techniques like **distance-based, density-based, or rank-based methods** to identify outliers.

Statistical Based Outlier Detection : Parametric

- **Detection of Univariate Outliers Based on Normal Distribution**
- A dataset with a single attribute is univariate.
- If we assume the data follows a normal distribution, we estimate parameters using maximum likelihood and identify low-probability points as outliers.
- **Example: Univariate Outlier Detection using Maximum Likelihood**
- **Given Data: A city's July temperature for 10 years (sorted):**
24.0°C, 28.9°C, 28.9°C, 29.0°C, 29.1°C, 29.1°C, 29.2°C, 29.2°C, 29.3°C, 29.4°C.

Statistical Based Outlier Detection

- assume that the average temperature follows a normal distribution, which is determined by two parameters:
- the **mean, μ** , and the standard **deviation, σ** .
- We can use the maximum likelihood method to estimate the parameters μ and σ .
- Maximum likelihood estimates are computed as:

$$\log L(\mu, \sigma) = -N \log(\sigma\sqrt{2\pi}) - \sum_{i=1}^N \frac{(x_i - \mu)^2}{2\sigma^2}$$

$$\hat{\mu} = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$
$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2.$$

Statistical Based Outlier Detection

$$\hat{\mu} = \frac{24.0 + 28.9 + 28.9 + 29.0 + 29.1 + 29.1 + 29.2 + 29.2 + 29.3 + 29.4}{10} = 28.61$$

$$\begin{aligned}\hat{\sigma}^2 = & ((24.1 - 28.61)^2 + (28.9 - 28.61)^2 + (28.9 - 28.61)^2 + (29.0 - 28.61)^2 \\ & + (29.1 - 28.61)^2 + (29.1 - 28.61)^2 + (29.2 - 28.61)^2 + (29.2 - 28.61)^2 \\ & + (29.3 - 28.61)^2 + (29.4 - 28.61)^2) / 10 \simeq 2.29.\end{aligned}$$

$$\hat{\sigma} = \sqrt{2.29} = 1.51$$

- The most deviating value, 24.0°C, is 4.61°C away from the estimated mean.
- So, 24.0°C is an outlier.

Statistical Based Outlier Detection

- Similarly Z-Score normalization also can be used to detect outlier.
- For any given data points. Compute mean and standard deviation.

$$\mu = \frac{\sum X}{N}$$
$$\sigma = \sqrt{\frac{\sum (X - \mu)^2}{N}}$$

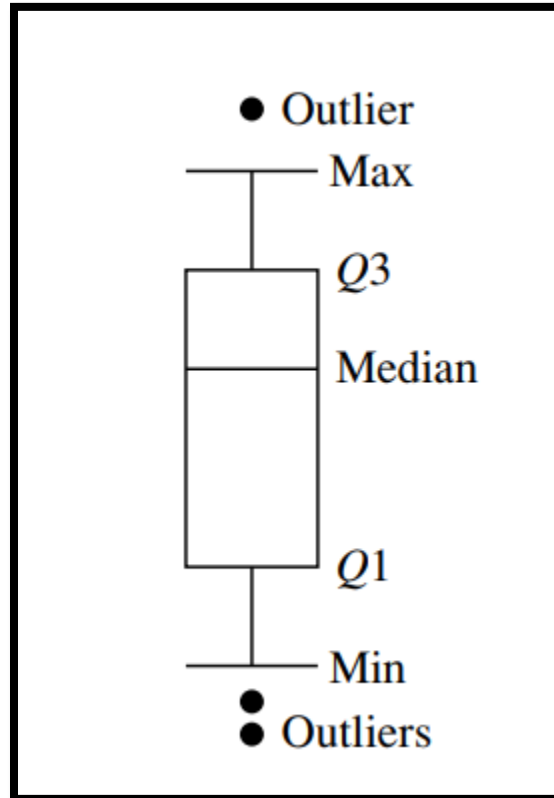
$$Z = \frac{x - \mu}{\sigma}$$

Data point is considered as an outlier if

$$|Z| > 3$$

Statistical Based Outlier Detection

- **Interpreting Box Plot for Outlier Analysis**



- A **box plot** consists of:
 - 1.**Minimum (Lower Whisker)**: Lowest value within the allowed range.
 - 2.**First Quartile (Q1)**: 25th percentile (lower boundary of the box).
 - 3.**Median (Q2)**: 50th percentile (middle value).
 - 4.**Third Quartile (Q3)**: 75th percentile (upper boundary of the box).
 - 5.**Maximum (Upper Whisker)**: Highest value within the allowed range.
 - 6.**Outliers**: Data points **outside the whiskers**.
- **Formula:**
 - **Interquartile Range (IQR) = $Q3 - Q1$**
 - **Lower Bound = $Q1 - 1.5 \times IQR$**
 - **Upper Bound = $Q3 + 1.5 \times IQR$**
- Any data point **outside these bounds** is an **outlier**.

- **Step : Example Calculation**
- **Given Dataset (Temperature in °C):**
 - [30,32,29,28,31,33,35,36,29,27,30,32,31,50]
- **Step: Arrange Data in Ascending Order**
 - [27,28,29,29,30,30,31,31,32,32,33,35,36,50]
- **Step: Find Q1, Q2 (Median), and Q3**
 - **Q1 (25th percentile):** Q1=29
 - **Q2 (Median or 50th percentile):** Q2=31
 - **Q3 (75th percentile) :** Q3=32

$$P = \frac{25}{100} \times (n + 1)$$

- **Step : Compute IQR**
- $\text{IQR} = Q3 - Q1 = 32 - 29 = 3$
- **Step : Compute Outlier Thresholds**
- **Lower Bound:** $Q1 - (1.5 \times \text{IQR}) = 29 - (1.5 \times 3) = 29 - 4.5 = 24.5$
- **Upper Bound:** $Q3 + (1.5 \times \text{IQR}) = 32 + (1.5 \times 3) = 32 + 4.5 = 36.5$
- **Step: Identify Outliers**
- Data **below 24.5** $\rightarrow$ **No values**
- Data **above 36.5** $\rightarrow$ **50°C is an outlier**

Statistical Based Outlier Detection

- **Detection of Multivariate Outliers**
- Multivariate data involves multiple attributes.
- Multivariate outlier detection using the Mahalanobis distance.
- For a multivariate data set, let $\bar{o}$ be the mean vector. For an object, o , in the data set, the Mahalanobis distance from o to $\bar{o}$ is:

$$MDist(o, \bar{o}) = (o - \bar{o})^T S^{-1} (o - \bar{o}),$$

- where S is the covariance matrix.

Statistical Based Outlier Detection

- Therefore, we can transform the multivariate outlier detection tasks as follows:

1. Calculate the mean vector from the multivariate data set.
2. For each object $\mathbf{o}$, calculate $MDist(\mathbf{o}, \bar{\mathbf{o}})$, the Mahalanobis distance from $\mathbf{o}$ to $\bar{\mathbf{o}}$.
3. Detect outliers in the transformed univariate data set, $\{MDist(\mathbf{o}, \bar{\mathbf{o}}) | \mathbf{o} \in D\}$.
4. If $MDist(\mathbf{o}, \bar{\mathbf{o}})$ is determined to be an outlier, then $\mathbf{o}$ is regarded as an outlier as well.

Statistical Based Outlier Detection

- Multivariate outlier detection using the χ^2 –statistic
- The χ^2 -statistic can also be used to capture multivariate outliers under the assumption of normal distribution. For an object, o, the χ^2 -statistic is :

$$\chi^2 = \sum_{i=1}^n \frac{(o_i - E_i)^2}{E_i},$$

- where o_i is the value of o on the i^{th} dimension, E_i is the mean of the i -dimension among all objects, and n is the dimensionality.
- If the χ^2 -statistic is large, the object is an outlier.

Distance-Based Outlier Detection

- Proximity-based approaches utilize a distance measure to determine the similarity between objects in a feature space.
- Objects that are significantly distant from others are considered outliers.
- These methods assume that the proximity of an outlier object to its nearest neighbors deviates significantly from the proximity of typical objects in the dataset.
- **Types of Proximity-Based Outlier Detection**
 - 1.Distance-Based Methods:** These methods evaluate the neighborhood of an object within a given radius. An object is classified as an outlier if its neighborhood does not contain enough points.
 - 2.Density-Based Methods:** These methods analyze the density of an object relative to its neighbors. If the density of an object is significantly lower than that of its neighbors, it is identified as an outlier.

Distance-Based Outlier Detection

Concept of Distance-Based Outliers

- A dataset **D** of objects is analyzed using a user-defined distance threshold **r** to define the neighborhood of an object.
- An object **o** is examined based on the number of other objects within its **r-neighborhood**.
- If most objects in **D** are far from **o** (i.e., not in its **r-neighborhood**), **o** is classified as an outlier.

Distance-Based Outlier Detection

Algorithm: Distance-based outlier detection.

Input:

- a set of objects $D = \{o_1, \dots, o_n\}$, threshold r ($r > 0$) and π ($0 < \pi \leq 1$);

Output: $DB(r, \pi)$ outliers in D .

Method:

```
for  $i = 1$  to  $n$  do
     $count \leftarrow 0$ 
    for  $j = 1$  to  $n$  do
        if  $i \neq j$  and  $dist(o_i, o_j) \leq r$  then
             $count \leftarrow count + 1$ 
            if  $count \geq \pi \cdot n$  then
                exit { $o_i$  cannot be a  $DB(r, \pi)$  outlier}
            endif
        endif
    endfor
    print  $o_i$  { $o_i$  is a  $DB(r, \pi)$  outlier according to (Eq. 12.10)}
endfor;
```

- let r ($r \geq 0$) be a distance threshold and π ($0 < \pi \leq 1$) be a fraction threshold.
- An object, o , is a $DB(r, \pi)$ -outlier if

$$\frac{\|\{o' \mid dist(o, o') \leq r\}\|}{\|D\|} \leq \pi$$

Figure Nested loop algorithm for $DB(r, \pi)$ -outlier detection.

Distance-Based Outlier Detection

Steps of the Algorithm

1. Input Parameters:

1. A dataset **D** containing **n** objects.
2. A distance threshold **r** that defines a neighborhood.
3. A fraction threshold **π** that determines the minimum number of neighbors an object must have to not be considered an outlier.

2. Processing Each Object:

1. Iterate over each object **oi** in the dataset.
2. Initialize a **count** variable to store the number of neighbors within distance **r**.

3. Checking Neighborhoods:

1. For each object **oi**, iterate through all other objects **oj** in the dataset.
2. If the distance between **oi** and **oj** is $\leq r$, increment the count.
3. If the count reaches $\pi \times n$, exit the loop (as **oi** is not an outlier).

4. Identifying Outliers:

1. If the count remains less than $\pi \times n$ after checking all objects, **oi** is classified as an outlier.

Point Name	X	Y
O1	1	1
O2	2	2
O3	2	3
O4	8	8
O5	9	9

Parameters:

- Distance threshold $r = 2.5$
- Fraction threshold $\pi = 0.4$
- Total objects $n = 5$
- Minimum required neighbors $\pi \times n = 0.4 \times 5 = 2$

Step-by-Step Execution

1.For o1:

1. Neighbors within $r = 2.5$: **o2**
2. Total count = **1** (less than 2) → **o1 is an outlier**

2.For o2:

1. Neighbors within $r = 2.5$: **o1, o3**
2. Total count = **2** (meets the threshold) → **o2 is not an outlier**

3.For o3:

1. Neighbors within $r = 2.5$: **o2**
2. Total count = **1** (less than 2) → **o3 is an outlier**

4.For o4:

1. Neighbors within $r = 2.5$: **o5**
2. Total count = **1** (less than 2) → **o4 is an outlier**

5.For o5:

1. Neighbors within $r = 2.5$: **o4**
2. Total count = **1** (less than 2) → **o5 is an outlier**

Final Outliers

o1, o3, o4, and o5 are outliers, while **o2** is not.

Distance-Based Outlier Detection

Strengths

- 1. Conceptually Simple:** The algorithm is easy to understand and implement.
- 2. Effective for Global Outliers:** It efficiently detects outliers that are far from dense clusters.
- 3. Parameter Control:** The r and π parameters allow flexibility in defining outliers.

Weaknesses

- 1. High Computational Cost:** The nested loop approach has a worst-case complexity of $O(n^2)$, making it inefficient for large datasets.
- 2. Sensitive to Parameter Choice:** The results heavily depend on the values of r and π , which must be tuned carefully.
- 3. Struggles with Varying Densities:** It does not perform well in datasets with varying densities, where some clusters may be sparse while others are dense.
- 4. Memory Intensive:** When dealing with large datasets, the approach requires a large memory space to store distance calculations.

Deviation-Based Outlier Detection

- Deviation-based outlier detection: It focuses on identifying outliers by looking at **how much a data point deviates from the normal pattern or trend of the dataset**, rather than using predefined statistical measures (like mean, standard deviation) or distance metrics (like Euclidean distance).

1.Does Not Rely on Statistical Distributions

1. Unlike Z-score or IQR methods, it does not assume that the data follows a normal or specific distribution.

2.Does Not Use Distance Metrics Directly

1. Unlike kNN-based or clustering-based approaches, deviation-based methods do not measure how far a point is from others in terms of numerical distance.

3.Focuses on Sequential or Structural Deviations

1. It looks at how data behaves in a sequence or structure rather than numerical deviations.

Deviation-Based Outlier Detection

Day	Temperature
1	30
2	31
3	30
4	29
5	30
6	31
7	30
8	29
9	45
10	30

- **Sequential Deviation**

- Sequential deviation looks for sudden, unexpected changes in a sequence of data. It is used in time-series data where values are expected to follow a trend.

- **Analysis:**

- Most of the temperatures are between **29°C and 31°C**.
- On **Day 9**, the temperature **jumps to 45°C**, which is a significant deviation from the normal range.
- This sudden spike is considered an **outlier**.

Deviation-Based Outlier Detection

Detection Approach:

- Compute the moving average:

- If we take a **5-day moving average**:

$$\text{Moving Average} = \frac{\sum_{i=t-2}^{t+2} \text{Temp}}{5}$$

- Before Day 9, the moving average is around **30°C**.
 - Day 9 has **45°C**, which **deviates significantly**.
- A deviation threshold (e.g., $\pm 5^{\circ}\text{C}$ from the moving average) can help flag such anomalies.

Outlier Analysis : Data Mining Applications

- Fraud Detection in **Healthcare Insurance** using Outlier Analysis.
- Healthcare insurance fraud includes fraudulent claims, overbilling, and fake treatments. Outlier detection helps identify suspicious activities that deviate significantly from normal claims.
- Example: A health insurance company wants to detect fraudulent claims in hospitalization expenses. Normally, hospital stays cost between Rs 5,000 and Rs. 50,000, but some claims exceed Rs 200,000, raising suspicion.

Steps for Outlier Detection in Healthcare Insurance Fraud

Step 1: Collect & Preprocess Data

- Gather historical claim data with attributes:
 - Claim ID
 - Patient Age
 - Hospital ID
 - Total Bill Amount
 - Number of Procedures
 - Length of Stay
 - Diagnosis Code
 - Insurance Policy Type
- Handle missing values and normalize data.

Step 2: Identify Outliers using Different Techniques

- 1. Statistical Methods (Z-Score & IQR Analysis)
- 2. Machine Learning Methods
- 3. Rule-Based Detection

Step 3: Investigate & Take Action

- Flagged claims go through manual review.
- Cross-check with patient records and medical history.
- Notify investigators for further validation.